Demonstration of Proposed Features: End-to-End Capabilities Showcase

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Module: `8.Project Demonstration`

Phase: Feature Showcase & Systems Demonstration

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1. Feature Showcase Overview

This document presents a structured feature-by-feature mapping between the project requirements and the live operational interfaces implemented across the NeuroClass edge platform.

NEUROCLASS CORE FEATURE MAP		
Feature ID & Name	Primary Interface / Endpoint	Core System Capability
F-01: Privacy Video HUD	Web HUD (`http://localhost:8000`)	Live feed with face privacy blur
F-02: 3-Way Enrollment	`[IDENTIFY / LINK] Identify / Link` on HUD Modal	Instant crop & biometric registry
F-03: Sustained Presence	Attendance Roster Drawer & DB	65% sliding window, walk-by drop
F-04: Head Pose Tracker	Visual Gaze Meter & Vector Overlay	3D Yaw/Pitch angle classification
F-05: Acoustic VAD Meter	Librosa Audio Panel on HUD	RMS energy, dialogue ratio (No STT
F-06: Multimodal Fusion	Composite Engagement Gauge	50/20/30 weighted index (0-100%)
F-07: Pedagogical Agent	Live Toast Nudges on HUD	Monologue/silence alert + cooldown
F-08: Streamlit Cockpit	Streamlit Dashboard (Port 8501)	Longitudinal analytics & CSV audit

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2. In-Depth Feature Demonstrations

The following sections provide deep-dive demonstrations of each core capability, illustrating both instructor and student interfaces alongside technical execution details.

Feature 1: Privacy-Preserving Video HUD

- **Operational Interface:** Web HUD (`frontend/index.html`) served at `http://127.0.0.1:8000`.
- **Demonstrated Capability:**
- Ingests video at 15-30 FPS from USB webcams, simulated feeds, or MP4 recordings.
- Automatically applies a ***31 \times 31*** Gaussian privacy mask over all detected faces in the display stream while passing unblurred normalized matrices to the internal feature extractor.
- Completely satisfies educational privacy regulations (GDPR/FERPA) by ensuring that identifiable raw face frames are never persisted to disk.

Feature 2: 3-Way Hybrid Biometric Enrollment

- **Operational Interface:** Live `[IDENTIFY / LINK] Identify / Link` button rendered beside active participant bounding boxes on the HUD.
- **Demonstrated Capability:**
- Evaluates unidentified participants as anonymous tracks (`TRK-001`).
- Allows the instructor to click `[IDENTIFY / LINK] Identify / Link` and assign official student identity (`STU-109`).
- Instantly saves a cropped portrait to `student_db/STU-109.jpg`, calculates the embedding, and updates all current and historical session attendance logs retroactively with zero frame drop.

Feature 3: Sustained Presence & Walk-By Filter

- **Operational Interface:** Live Attendance Summary Table on the Web HUD and `/api/attendance/summary`.
- **Demonstrated Capability:**
- Classifies visitors visible for ***<15*** continuous seconds as transient walk-bys, discarding them from the attendance record.

• Tracks continuous presence duration for enrolled participants. When active time exceeds ***\ge 65\%*** of the session window, the student's status permanently changes from `UNVERIFIED` to `PRESENT`.

Feature 4: Visual Attention & Head Pose Gaze Estimation

• **Operational Interface:** Visual Attention Vector Gauge on the Web HUD.

• **Demonstrated Capability:**

• Computes 3D head rotation angles (***\text{Yaw}*** and ***\text{Pitch}***).

• Categorizes students as **Attentive** when ***|\text{Yaw}| \le 25^\circ \land |\text{Pitch}| \le 20^\circ***.

• Categorizes students looking down at smartphones or turning around as **Distracted**.

Feature 5: Privacy-Preserving Acoustic VAD & Dynamics

• **Operational Interface:** Audio Dynamics Monitor on the Web HUD (`/ws/audio`).

• **Demonstrated Capability:**

• Standardizes raw 16 kHz microphone buffers into RMS decibel levels and Voice Activity Detection (VAD) ratios.

• **Zero Speech-to-Text Transcription:** The system analyzes audio dynamics and conversational ratios without decoding or storing spoken words.

Feature 6: Multimodal Temporal Fusion

• **Operational Interface:** Composite Engagement Speedometer Gauge on the Web HUD.

• **Demonstrated Capability:**

• Merges visual gaze orientation (***50\%***), physical motion balance (***20\%***), and acoustic dialogue (***30\%***) across a synchronized 5-second sliding window.

• Outputs an explainable composite Engagement Index (***0-100\%***) refreshed continuously.

Feature 7: Autonomous Pedagogical Reasoning Agent

• **Operational Interface:** Floating Real-Time Alert HUD.

• **Demonstrated Capability:**

• Detects continuous instructor monologues exceeding 12 minutes or classroom silence exceeding 6 minutes.

• Fires an unobtrusive pedagogical toast recommendation.

• Locks further toast triggers for 45 seconds to prevent alert fatigue.

Feature 8: Streamlit Analytics Cockpit & CSV Audit Export

• **Operational Interface:** Streamlit Dashboard at `http://127.0.0.1:8501`.

• **Demonstrated Capability:**

• Generates chronological session attendance heatmaps, focus zone distributions, and student attentional percentiles.

• Exports a certified, tamper-evident official attendance CSV with student IDs, timestamps, presence minutes, and attendance marks.